

WGS FINAL REPORT (SYNTHETIC)

Patient ID	SIM-MERRF-01	Ordering Physician	Dr. [Ordering Physician]	Specimen	Simulated peripheral blood
Sex	Male	Account#	SIM-PIPE	Collected	2026-07-15
Date of Birth	XX/XX/XXXX (synthetic)	Source	Pipeline v1.1 test dataset	Received	2026-07-15
Indication	Simulated: suspected mitochondrial myopathy	Case ID	CS-MERRF_INXXXX	Reported	2026-07-15

TEST RESULT: A pathogenic mitochondrial point mutation, **m.8344A>G** in **MT-TK**, was detected at 78% heteroplasmy in this simulated case. Pathogenic MT-TK variants are associated with **MERRF syndrome** (Myoclonic Epilepsy with Ragged Red Fibers) [MIM: 545000], characterized by myoclonus, generalized epilepsy, cerebellar ataxia, myopathy, and ragged red fibers on muscle biopsy.

PATIENT PHENOTYPE (SIMULATED)

Simulated phenotype terms used to drive prioritization: myoclonus, seizures, ataxia, muscle weakness.

PRIMARY FINDINGS — Variants in genes associated with patient's simulated phenotype

Confirmation Status	Gene / Locus	Condition	Chrom : Coordinates	Variant	Zygosity	Inheritance	Classification
Confirmed	MT-TK	MERRF Syndrome	chrM:8,344	m.8344A>G	Heteroplasmic (78%)	Mitochondrial	Pathogenic
Pending Review	MT-TK	MERRF Syndrome (candidate)	chrM:8,356	m.8356T>C	Heteroplasmic (6%)	Mitochondrial	Discordant — ClinVar VUS / InterVar Likely Pathogenic

INTERPRETATION

Variant Information

The classic MERRF hotspot m.8344A>G was detected at 78% heteroplasmy, well above the 10% pathogenic-heteroplasmy threshold used by this pipeline, and is classified Pathogenic concordantly by both ClinVar and InterVar. A second candidate variant at the same tRNA locus, m.8356T>C, was also detected but at only 6% heteroplasmy: ClinVar lists it as Uncertain Significance while InterVar's rule engine leans toward Likely Pathogenic on the basis of its known-hotspot position. This produces a **discordant call** that the pipeline flags for manual review rather than auto-resolving. A third variant at this locus, present at 100% (i.e. homoplasmic), is a common benign haplogroup marker and does not affect interpretation.

Gene Information

MT-TK encodes the mitochondrial tRNA for lysine. Pathogenic variants disrupt mitochondrial protein translation, impairing oxidative phosphorylation most prominently in high-energy-demand tissues (skeletal muscle, brain), which underlies MERRF's myopathic and neurological features. Heteroplasmy level (the proportion of mutant vs wild-type mitochondrial genomes) is a major determinant of phenotype severity and is reported quantitatively for every mitochondrial call.

REFERENCES

Shoffner JM, et al. Cell. 1990. Myoclonic epilepsy and ragged-red fiber disease (MERRF) is associated with a mitochondrial DNA tRNA(Lys) mutation.

DiMauro S, Hirano M. GeneReviews. Updated periodically. MERRF.

Richards S, et al. Genetics in Medicine. 2015. Standards and guidelines for the interpretation of sequence variants (PMID: 25741868).

RECOMMENDATIONS

- Clinical correlation with the patient's phenotype is recommended.
- Because this dataset is synthetic pipeline-validation data, no clinical action should be taken on it.
- Genetic counseling is recommended in real cases to discuss implications for the patient and family.
- Orthogonal confirmation (e.g. Sanger sequencing, long-read sequencing, or inverse-PCR where applicable) is recommended before clinical sign-out of any pathogenic or likely-pathogenic call.
- Periodic re-evaluation of variant classification is appropriate as knowledge and databases evolve.

TEST METHODOLOGY

This report was produced by the RareDiseasePipeline Nextflow/Colab workflow, a synthetic-data validation harness modeled on production rare-disease WGS interpretation pipelines. Variants shown here were generated by the pipeline's own per-disease test-data generator (Section 12.1 of the pipeline notebook), then passed through the same region-filtering, annotation, disease-specific-branch, prioritization, and ACMG classification steps used for real inputs: VEP and SnpEff for functional consequence, ClinVar/SnpSift for clinical significance, gnomAD for population frequency, SpliceAI/REVEL/AlphaMissense/CADD for in-silico pathogenicity prediction, InterVar for ACMG/AMP classification, ClassifyCNV and SvAnna for structural-variant interpretation, ClinGen for gene dosage sensitivity, and ExpansionHunter/TRGT for repeat-expansion sizing. Variant classification follows the ACMG/AMP 2015 sequence-variant interpretation guidelines (Richards et al. 2015).

TEST LIMITATIONS

This is a synthetic dataset generated for pipeline validation and does not represent a real patient or real sequencing data. As with any WGS-based test, full genome coverage is not achievable in repetitive or duplicated regions, and certain variant classes (e.g. deep intronic variants, some repeat expansions, complex structural rearrangements) may not be reliably detected. Structural-variant and inversion calls that are reported by only one tool in the chain are flagged as candidates and require orthogonal confirmation before they can be considered clinically actionable.

REGULATORY DISCLOSURES

This report was generated by a personal/educational bioinformatics pipeline (RareDiseasePipeline) built for pipeline development and validation purposes. It has not been developed, validated, or reviewed by any accredited clinical laboratory, has not been cleared or approved by the FDA, and is not a CLIA- or CAP-certified clinical test result. It must not be used for patient care, diagnosis, or any clinical decision-making.

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Pipeline Developer / QA Lead

[Reviewer Name]
Secondary Reviewer (placeholder)

Test results reviewed and approved by: [Reviewer Name] — placeholder signature, synthetic report.

Patient	Case ID	DOB	Sex	Report Date	Page
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